ACDT PROJECT 1 / EXECUTIVE SUMMARY

BEYOND THE RATINGS

Do better-reviewed movies earn more worldwide revenue?

2022-2025 | 385 films | Metacritic Metascore | Worldwide gross in USD

RESEARCH QUESTION & DESIGN

We tested whether higher critic Metascore is associated with greater worldwide box-office gross after adjustment for production budget, release year, and primary genre. The 385-film sample includes Action, Comedy, Drama and Horror. Gross and budget enter the analysis on the natural-log scale.

KEY FINDINGS

Simple OLS

-9.1%

p = 0.205

Multiple OLS

+22.9%

p = 0.0008

RF-DML

+24.3%

p ≈ 0.000085

All percentage figures refer to the modeled difference in worldwide gross associated with a 10-point Metascore increase. The multiple OLS model has $R^2 = 0.582$; the genre-interaction Wald test has p = 0.0493. In the interaction model, the Action estimate is +45.8% per 10 points, but other genre slopes are imprecise.

MACHINE LEARNING & SYSTEM MAPS

Random Forest performs slightly better than linear regression in 5-fold prediction validation (mean R^2 0.585 vs 0.547). Budget drives most of the model's predictive importance. Map A presents regression-supported associations; Map B adds plausible but unmeasured industry factors; Map C separates Random Forest prediction from RF-based DML adjustment.

LIMITATIONS & DECISION IMPLICATIONS

The dataset is observational and selected for complete reported budgets, gross and ratings. The 2025 coverage is partial; genre coding and sample selection influence moderation results. Reviews, marketing, franchises, distribution and release strategy may jointly respond to film success. These results are associations, not demonstrated causal effects. Industry decisions should not rely on ratings alone.

Source: group-supplied cleaned IMDb-derived Kaggle dataset (Raed Addala), ACDT_Project1_USD_Dataset_Updated notebook. Inference: heteroskedasticity-robust HC3 OLS; cross-fitted partially linear RF-DML; scikit-learn cross-validation. See analysis notebook and System_Map_Justification.md.